



Modeling cell trajectories

Applications in ML

- Time-series alignment via OT geodesics.
- Interpolation in generative models (e.g. WGAN latent paths).
- **Flow matching** → replace naive interpolation with OT interpolation
- Diffusion-based generative models → Schrödinger bridges (next lecture).
- **Modeling cell trajectories**

Application: Flow Matching

Flow Straight and Fast:
Learning to Generate and Transfer Data with Rectified Flow

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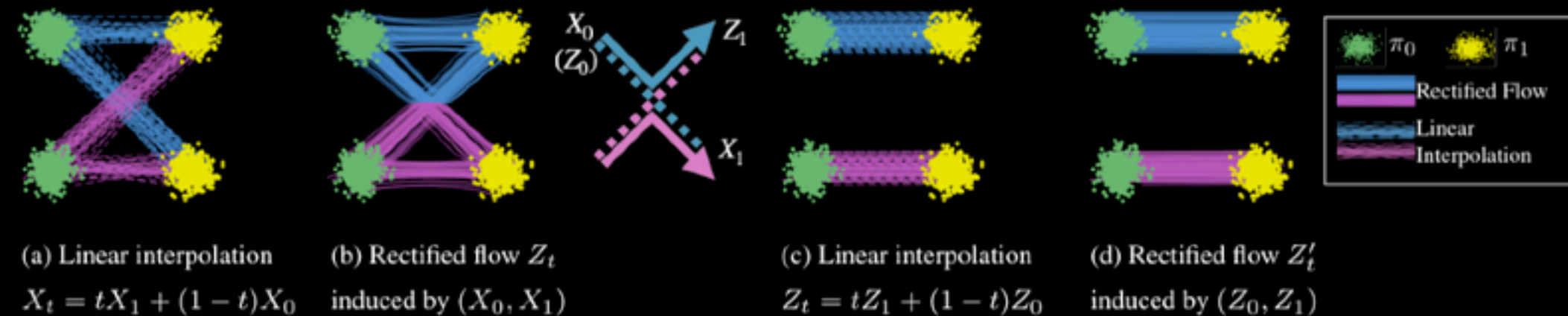


Figure 2: (a) Linear interpolation of data input $(X_0, X_1) \sim \pi_0 \times \pi_1$. (b) The rectified flow Z_t induced by (X_0, X_1) ; the trajectories are “rewired” at the intersection points to avoid the crossing. (c) The linear interpolation of the end points (Z_0, Z_1) of flow Z_t . (d) The rectified flow induced from (Z_0, Z_1) , which follows straight paths.

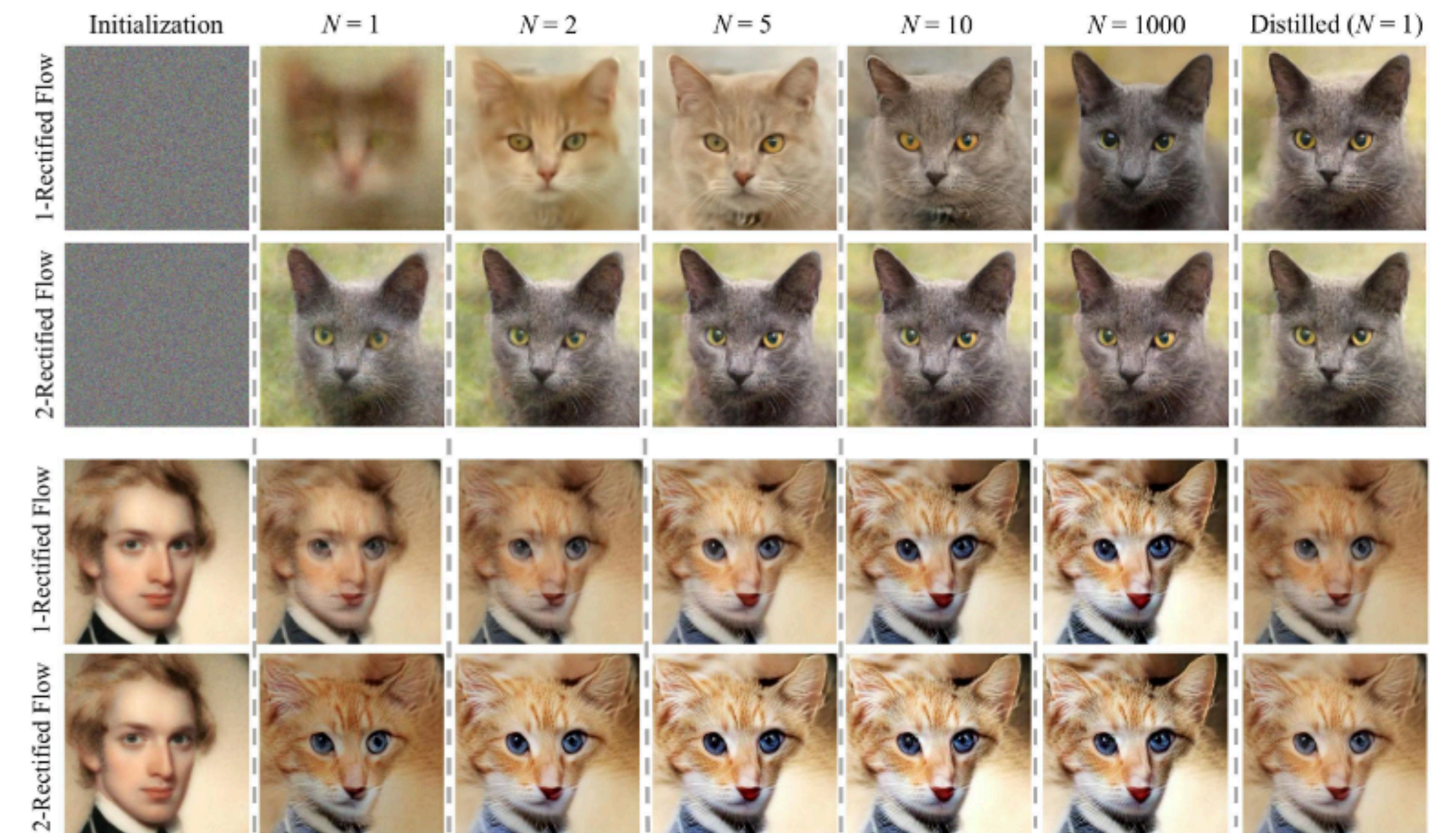


Figure 1: The trajectories of rectified flows for image generation (π_0 : standard Gaussian noise, π_1 : cat faces, top two rows), and image transfer between human and cat faces (π_0 : human faces, π_1 : cat faces, bottom two rows), when simulated using Euler method with step size $1/N$ for N steps. The first rectified flow induced from the training data (*1-rectified flow*) yields good results with a very small number (e.g., ≥ 2) of steps; the straightened reflow induced from *1-rectified flow* (denoted as *2-rectified flow*) has nearly straight line trajectories and yield good results even with one discretization step.